

FINITE-REFERENCE ERROR BOUNDS FOR RECURRENT EVALUATORS IN POLICY IMPROVEMENT

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ABSTRACT

Recurrent evaluators used in policy improvement stop after a finite number of internal steps, even when their computation remains transient. We compare the deployed depth- n value with a deeper finite endpoint at depth m . The value error is bounded by the endpoint discrepancy plus the reference endpoint’s self-bootstrap Bellman residual divided by $1 - \gamma^K$. We carry this radius through exact advantage centering into conservative policy improvement with signed evaluation error, and separate propagated target-network lag from deployment mismatch. For persistent latents, the clock-complete augmented state and one shared frozen recurrent map keep both endpoints in the same MDP. An exact calculation on two nonabsorbing states plus a shared absorber gives a nonvacuous value radius and a positive CPI lower bound. The guarantees concern one candidate against a frozen base at a fixed parameter snapshot.

1 INTRODUCTION

Tiny Recursive Models (TRMs) refine an internal representation before producing an answer (Jolicoeur-Martineau, 2025). Unrolled Policy Iteration with TRMs (UPI-TRM) uses this inner computation to evaluate plans in a checker-rewarded edit MDP: a value head supplies approximate advantages, a candidate policy chooses edits, and each edit advances the environment.

The evaluator is deployed after n recurrent steps, even when its internal computation is still changing. Comparing its output U_n with a converged recurrence requires a latent fixed point and control of the distance to it. Useful learned computation can instead be transient. We bound the error of the deployed finite-depth evaluator by comparing it with another finite endpoint, without assuming recurrent convergence.

A finite reference. Fix the policy π and compare U_n with the same evaluator at a deeper finite depth $m > n$. The identity

$$U_n - V^\pi = (U_n - U_m) + (U_m - V^\pi)$$

separates sensitivity to additional computation from reference accuracy. Standard Bellman-residual analysis bounds the latter by $\|U_m - \mathcal{T}_K^\pi U_m\|_\infty / (1 - \gamma^K)$, using a K -step current-policy backup that self-bootstraps from U_m . Theorem 5.1 gives the full conditions. A deeper reference helps only when its residual compensates for the endpoint discrepancy; depth alone does not guarantee a tighter bound.

Two depths in one MDP. The Bellman argument treats the endpoints as bounded functions. Recurrence matters when interpreting them as two computation depths of the same deployed evaluator. With persistent latents, the depth- n map determines what is carried into the successor state. Substituting a depth- m transition would change the MDP. We instead keep the policy and transition kernel fixed and vary only the endpoint used for evaluation.

From evaluation to improvement. The value radius controls advantage error, which enters an exact probability-mixture CPI bound (Kakade & Langford, 2002). Exact centering removes the

base-policy component of evaluation error; signed defects retain information lost by absolute worst-case bounds. Target-network lag and approximate deployment are separate corrections.

The contributions are a finite-reference recurrent certificate (Theorem 5.1), its advantage/CPI and target-network connection (Proposition 6.1 and Theorem 7.3), and persistent-state Markovization with a shared frozen map (Corollary 8.1). An exact finite-MDP calculation and an implementation-level account of targets and replay complete the analysis. The proofs use established residual and CPI arguments at one fixed parameter snapshot.

2 RELATED WORK

TRMs and HRMs use recurrent latent refinement for structured prediction (Jolicoeur-Martineau, 2025; Wang et al., 2025); deep equilibrium models use an equilibrium representation (Bai et al., 2019). Here the evaluator is read at finite depth, with contraction needed only for the optional fixed-point specialization. Classical approximate dynamic programming controls value error and its propagation through policy or value updates (Bertsekas & Tsitsiklis, 1996; Munos, 2003; Munos & Szepesvári, 2008; Scherrer, 2014). Our primary result instead uses a fixed-policy, population sup-norm residual; Appendix G.2 states the coverage condition for its weighted-norm version.

CPI and safe policy iteration control occupancy shift and approximate improvement (Kakade & Langford, 2002; Kakade, 2003; Pirota et al., 2013; Metelli et al., 2021). We supply the evaluation term for one exact probability mixture from two finite recurrent endpoints. Learning to search casts structured prediction as sequential decisions (Daumé III et al., 2009; Ross et al., 2011; Chang et al., 2015); verifiable-reward systems use automatic correctness signals (DeepSeek-AI, 2025). Our inner depth n is computation within one evaluation, distinct from planning over transitions (Tamar et al., 2016) or the Bellman horizon K .

3 BACKGROUND AND SETUP

The outer edit MDP. An edit a_t changes plan y_t for a fixed instance x . With h_t edits remaining, the episodic state and nonterminal successor are

$$s_t = (x, y_t, h_t), \quad s_{t+1} = (x, \text{edit}(y_t, a_t; x), h_t - 1).$$

We call a state clock-complete when it carries the remaining budget h . Solving, terminal STOP, or budget exhaustion sends the process to the shared absorber s_{abs} . A bounded checker score $c(x, y)$ supplies reward, optionally shaped as $r = r_0 + \gamma\Phi(s') - \Phi(s)$ with $\Phi = c$ (Ng et al., 1999). We maximize discounted return $\eta(\pi)$ from ρ , with $\gamma \in [0, 1)$. The shaped boundary is $V^\pi(s_{\text{abs}}) = b = -C_{\text{max}}$; folding its tail into the terminal reward with zero terminal bootstrap preserves returns (Appendices A.2 and H).

The inner evaluator. A recurrent map refines a latent while the instance and current plan remain fixed. Starting from $z^{(0)} = z_{\text{init}}(x, y)$ in episodic mode,

$$z^{(k+1)} = \tilde{f}_\theta(z^{(k)}, y, x), \quad U_n(s) = V_\psi(z^{(n)}, x, y).$$

Depth n counts internal computation; the Bellman horizon $K \geq 1$ counts environment transitions. Here $\tilde{f}_\theta = P_{\text{rec}} \circ f_\theta$, where $P_{\text{rec}} = \Pi_R$ is Euclidean projection onto a radius- $R > 0$ ball, or the identity when disabled. The projection is defined in Eq. (39). The implemented head reads (z, x, y) , not h , but rewards, termination, and successors need the clock.

Reset or carried latent state. Resetting z at each edit makes (x, y, h) Markov. Carrying it requires $\bar{s} = (x, y, z, h)$, since different latents can produce different decisions. The state stores the *pre-unroll* latent; deployment computes $F_n(x, y, z) = \tilde{f}_\theta^{\circ n}(z, y, x)$, uses its output for action selection, and carries it only on nonterminal transitions. Both CPI policies share this frozen map. The comparison endpoint $\tilde{U}_m$ starts from stored z without changing F_n , action sampling, or successors.

Values and domains. Fix the MDP, recurrence, initialization, value head, and current/candidate policies. State/action spaces are measurable, policies are stochastic kernels, and values are bounded

Algorithm 1 One fixed-base UPI-TRM proposal on the clock-complete edit MDP

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- 1: Freeze the base policy and recurrent map; sample x , initialize y_0 , and set $h_0 = T \geq 1$.
 - 2: Use the mode-appropriate state s_t : (x, y_t, h_t) in episodic mode, (x, y_t, z_t, h_t) in persistent mode; initialize persistent z_0 .
 - 3: **for** $t = 0, \dots, T - 1$ **do**
 - 4: Use $z_t = z_{\text{init}}(x, y_t)$ in episodic mode or stored z_t in persistent mode; compute $z_t^{(n)} = \tilde{f}_{\theta}^{\circ n}(z_t, y_t, x)$ and $V_{\psi}(z_t^{(n)}, x, y_t)$.
 - 5: Sample $a_t \sim \pi_{\phi}(\cdot \mid y_t, z_t^{(n)}, x)$; observe r_t, y_{t+1}, d_t and set $h_{t+1} = h_t - 1$.
 - 6: Set $s_{t+1} = s_{\text{abs}}$ if d_t or $h_{t+1} = 0$; otherwise construct the successor, carrying $z_t^{(n)}$ only in persistent mode.
 - 7: Store (s_t, a_t, r_t, s_{t+1}) ; stop collection immediately if $s_{t+1} = s_{\text{abs}}$.
 - 8: **end for**
 - 9: Form exact- K or terminal folded targets and fit the mode-appropriate value head.
 - 10: Compute advantages by Eq. (2), conditional on the complete state.
 - 11: Fit one candidate policy head while the base policy and recurrent map remain frozen.
 - 12: Form the exact probability mixture $\pi_{\alpha} = (1 - \alpha)\pi_{\phi} + \alpha\pi_{\text{cand}}$.
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measurable functions. On $\mathcal{C}$, $\|f\|_{\infty} = \sup_{s \in \mathcal{C}} |f(s)|$ is the ordinary pointwise supremum. Every endpoint equals b at the absorber; latent paths are defined only elsewhere. Section 5 introduces the advantage-evaluation domain; Appendix A.1 gives all domain definitions.

4 UPI-TRM ALGORITHM

Algorithm 1 freezes the base edit policy and recurrent map before collection. Checker rewards and K -step targets fit the value head; U_n -derived centered advantages fit one candidate policy head. The proposal mixes action probabilities: $\pi_{\alpha} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$. Optimizer steps fit this proposal without replacing the base.

For a bounded value function V , define the exact one-step estimator

$$\hat{Q}_V(s, a) = \mathbb{E}_{s'}[r(s, a, s') + \gamma V(s')], \quad (1)$$

$$\hat{A}_V(s, a) = \hat{Q}_V(s, a) - \mathbb{E}_{b \sim \pi(\cdot \mid s)}[\hat{Q}_V(s, b)]. \quad (2)$$

The candidate uses $V = U_n$. For persistent latents, condition on the full $\bar{s}$. The theorems analyze the fitted heads at one frozen snapshot. Approximating the mixture by a single network requires the separate deployment correction in Section 8.

Only nonterminal transitions carry a latent; every terminal cause ends collection after its final record. Appendix H gives folded targets and replay details. Exact baseline enumeration ensures statewise centering; approximate integration instead contributes the signed defect in Theorem 7.3. One depth- n unroll per action term over T states costs $O(T|A|n)$ recurrent-map applications. Interpolating weights, logits, or hidden states, or distilling the mixture, generally changes its law.

5 FINITE-REFERENCE EVALUATION AND ADVANTAGE ERROR

We first use episodic notation; Section 8 applies the same argument on the fixed augmented MDP.

5.1 THE FINITE-REFERENCE VALUE CERTIFICATE

The triangle inequality splits $\|U_n - V^{\pi}\|_{\infty}$ into endpoint discrepancy and reference error. Bellman contraction controls the latter through its self-bootstrap residual (Puterman, 1994; Sutton & Barto, 2018).

For later advantage estimation, let $\mathcal{C}_{\pi}$ carry the initial law and be invariant under π . Write $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ for a measurable set containing $\mathcal{C}_{\pi}$, the absorber, and successors of one current or candidate action, followed by any current-policy continuation (Definition A.3). Define $\mathcal{C}^{\circ} = \mathcal{C} \setminus \{s_{\text{abs}}\}$ and

$\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} = \sup_{s \in \mathcal{C}} |f(s)|$. On this invariant domain the residual bound is

$$\|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{1}{1 - \gamma^K} \|V - \mathcal{T}_K^\pi V\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (3)$$

Apply this estimate to $V = U_m$ and add the endpoint discrepancy:

Theorem 5.1 (Finite-reference bound). *Let $\mathcal{C}$ be a measurable state set with its trace σ -algebra, let P_π be a Markov kernel satisfying $P_\pi(\mathcal{C} \mid s) = 1$ for every $s \in \mathcal{C}$, and fix $\gamma \in [0, 1)$ and an integer $K \geq 1$. If an initial law is used, assume $\rho(\mathcal{C}) = 1$. Assume the one-step reward is uniformly bounded and its fixed-policy conditional mean $r^\pi(s) := \mathbb{E}_s^\pi[r_0]$ is measurable on $\mathcal{C}$. Then the K -step expected reward*

$$r_K^\pi(s) := \mathbb{E}_s^\pi \left[\sum_{i=0}^{K-1} \gamma^i r_i \right] \quad (4)$$

is bounded and measurable on $\mathcal{C}$. On the Banach space $B_b(\mathcal{C})$ of bounded measurable functions with the ordinary pointwise sup norm, define

$$(\mathcal{T}_K^\pi V)(s) := r_K^\pi(s) + \gamma^K (P_\pi^K V)(s). \quad (5)$$

Then $\mathcal{T}_K^\pi : B_b(\mathcal{C}) \rightarrow B_b(\mathcal{C})$ is a γ^K -contraction and has a unique fixed point V^π . For arbitrary endpoint functions $U_n, U_m \in B_b(\mathcal{C})$ and finite integers $0 \leq n < m$,

$$\|U_n - V^\pi\|_\infty \leq \underbrace{\|U_n - U_m\|_\infty}_{\text{finite-depth discrepancy}} + \underbrace{\frac{\|U_m - \mathcal{T}_K^\pi U_m\|_\infty}{1 - \gamma^K}}_{\text{reference-depth residual}}. \quad (6)$$

The discrepancy measures sensitivity to additional computation; the residual measures agreement with the frozen policy’s value equation. Both endpoints use the same policy and kernel as V^π . Appendix F.1 gives the proof and a nonconvergent endpoint example; Corollary F.1 gives the latent-path inequalities. Uniform recurrent contraction gives the optional geometric L_z^n limit in Appendix G.1. The weighted- L_p version in Appendix G.2 requires explicit domination of the fixed-policy occupancy by the reference measure.

5.2 FROM VALUE ERROR TO CANDIDATE ADVANTAGE ERROR

The action value and baseline each contribute at most γE from successor value error E , giving $2\gamma E$. Policy overlap sharpens the candidate-specific bound. Define

$$\tau(s) := \text{TV}(\pi_{\text{cand}}(\cdot \mid s), \pi(\cdot \mid s)), \quad \tau_\infty := \sup_{s \in \mathcal{C}_\pi} \tau(s) \leq 1. \quad (7)$$

We use $b(s) = \mathbb{E}_{\pi_{\text{cand}}}[\hat{A}(s, a) - A^\pi(s, a)]$ for the signed candidate defect, and $\varepsilon_{A, \text{cand}} = \sup_{s \in \mathcal{C}_\pi} |b(s)|$. The function $b(s)$ is distinct from the scalar absorbing boundary b . The normalized discounted occupancy is $d_\pi = (1 - \gamma) \sum_{t \geq 0} \gamma^t \rho P_\pi^t$.

Value error is controlled on $\mathcal{C}$, advantage error on $\mathcal{C}_\pi$. The check is post-candidate because $\mathcal{C}$ contains candidate successors.

Corollary 5.2 (Advantage-error bridge). *Under the hypotheses of Theorem 5.1, define*

$$B_{n,m} := \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (8)$$

Let $\hat{A} = \hat{A}_{U_n}$ be the exact one-step centered estimator of Eq. (2), with the transition expectation evaluated exactly (or applied to a deterministic transition) and the baseline computed exactly under π . Then Assumption 7.1 holds with the valid choice

$$\varepsilon_A = 2\gamma B_{n,m}. \quad (9)$$

The reference assesses the depth- n estimator without changing its action selection. Exact centering gives the sharper candidate-average bound:

Corollary 5.3 (Candidate evaluation bias from policy overlap). *Let $\hat{A}_V$ be the exactly centered one-step estimator, $e = V - V^\pi$, and $\delta_V = \|e\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Use τ and τ_∞ from Eq. (7). Then, for every $s \in \mathcal{C}_\pi$,*

$$\left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}_V(s, a) - A^\pi(s, a)] \right| \leq 2\gamma\delta_V\tau(s), \quad (10)$$

and hence

$$\varepsilon_{A, \text{cand}} \leq 2\gamma\delta_V\tau_\infty, \quad (11)$$

$$\mathbb{E}_{s \sim d_\pi} |b(s)| \leq 2\gamma\delta_V \mathbb{E}_{s \sim d_\pi} \tau(s). \quad (12)$$

For an exactly centered K -step estimator whose intervening actions follow π , replace γ by γ^K . The fallback $2\gamma\delta_V$ follows from $\tau \leq 1$.

Proof in Appendix F.4.

Exact centering cancels the shared part of the two action laws, so $2\gamma\delta_V\tau(s)$ can be much smaller than the worst-case $2\gamma\delta_V$, and is zero when the policies agree. Section 7 uses this candidate bias in an improvement bound. In particular, under the exact-mixture and centering hypotheses of Theorem 7.3, the finite-reference radius gives

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1-\gamma} B_{n,m} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (13)$$

Here $\hat{L}_\pi$ is the estimated CPI surrogate and ε_{CPI} bounds the magnitude of the candidate’s expected true advantage on the mixture domain; both are defined in Section 7. For centered K -step estimators the advantage factor becomes $2\gamma^K$; other estimators, including GAE, need their own error bound (Remark F.3).

6 TARGET-NETWORK RESIDUALS

Value regression fits U_n to a target network $\bar{V}$, whereas the certificate needs a backup of the evaluated function itself. For a frozen current-policy trajectory, the target is $G^{(K)} = \sum_{k < K} \gamma^k r_k + \gamma^K \bar{V}(S_K)$, with loss $\mathcal{L}_{\text{val}} = \mathbb{E}_{S_0 \sim \mu} [(U_n(S_0) - \text{sg}(G^{(K)}))^2]$. Here μ selects starting states; the conditional trajectory law is π . Even zero target residual can leave a self-bootstrap residual when the networks differ. The correction is propagated target lag:

Proposition 6.1 (Target-network residual). *Let $\mathcal{E}$ be a measurable P_π -invariant state domain. Assume $r_K^\pi \in B_b(\mathcal{E})$ and $U_q, \bar{V} \in B_b(\mathcal{E})$ for a finite depth $q \geq 0$, and suppose the conditional K -step target trajectory follows the frozen policy π . Define*

$$\varepsilon_{\text{targ}, q} := \|U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})\|_{\infty, \mathcal{E}}. \quad (14)$$

Retain the propagated target lag by defining

$$R_{\text{targ}, q}^P := \varepsilon_{\text{targ}, q} + \gamma^K \|P_\pi^K (\bar{V} - U_q)\|_{\infty, \mathcal{E}}. \quad (15)$$

Then the self-bootstrap residual satisfies

$$\|U_q - \mathcal{T}_K^\pi U_q\|_{\infty, \mathcal{E}} \leq R_{\text{targ}, q}^P. \quad (16)$$

For the formal absorbing representation, require $U_q(s_{\text{abs}}) = \bar{V}(s_{\text{abs}}) = b$ and use the same absorbing reward and boundary in both population operators. The equivalent folded implementation may instead fold γb into the terminal-transition reward and mask the terminal bootstrap to zero, but it must not add the absorbing tail again.

Proof in Appendix F.5. The ordinary sup-lag form $R_{\text{targ}, q}^{\text{sup}} := \varepsilon_{\text{targ}, q} + \gamma^K \|\bar{V} - U_q\|_{\infty, \mathcal{E}}$ is the non-expansive fallback: $R_{\text{targ}, q}^P \leq R_{\text{targ}, q}^{\text{sup}}$. Corollary D.1 gives the value/CPI substitutions. Population mean-square regression controls an $L^2(\mu)$ conditional residual.

7 CONSERVATIVE POLICY IMPROVEMENT

The exact mixture $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ can select its base or candidate component with a Bernoulli draw at each decision. CPI corrects the base-policy surrogate for advantage error and occupancy shift (Kakade & Langford, 2002); we keep signed error and bound shift using this selector. Error uses $\mathcal{C}_\pi$; occupancy needs $\mathcal{C}_{\text{mix}}$ to contain full trajectories under both policies, not just one deviation (Definition A.4). The uniform-error assumption below suffices for the scalar specialization but is not needed for the signed bound.

Assumption 7.1 (Uniform advantage error on $\mathcal{C}_\pi$). For some $\varepsilon_A \geq 0$:

$$|\hat{A}(s, a) - A^\pi(s, a)| \leq \varepsilon_A \quad (17)$$

for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a .

Write $d_\pi = (1 - \gamma) \sum_{t \geq 0} \gamma^t \rho P_\pi^t$ and define the true and estimated CPI surrogates

$$\begin{aligned} L_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [A^\pi(s, a)], \\ \hat{L}_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [\hat{A}(s, a)]. \end{aligned}$$

Lemma 7.2 (Exact-mixture occupancy). *Let π and π_{cand} act in the same MDP from the same initial law, and let π_α be their exact pointwise probability mixture. For $\alpha \in [0, 1]$,*

$$\text{TV}(d_{\pi_\alpha}, d_\pi) \leq \frac{\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (18)$$

For finite or countable state spaces, with $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$,

$$\|d_{\pi_\alpha} - d_\pi\|_1 \leq \frac{2\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (19)$$

Proof in Appendix F.6.

The order- α occupancy shift multiplies the candidate's order- α surrogate contribution. The resulting correction is $O(\alpha^2)$ for a fixed domain and advantage span. The evaluation correction combines base-policy centering defect $c(s)$ and candidate bias $b(s)$ under d_π , the surrogate's occupancy.

Theorem 7.3 (CPI with signed evaluation error). *Fix one discounted MDP with bounded rewards and measurable current and candidate policy kernels. Let $\alpha \in [0, 1]$ and let $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ be their exact pointwise probability-space mixture, with no parameter interpolation, hidden-state interpolation, logit interpolation, or distillation. Let $\mathcal{C}_{\text{mix}}$ be measurable with $\rho(\mathcal{C}_{\text{mix}}) = 1$ and invariant under both π and π_α , as in Definition A.4. Assume that a measurable π -invariant domain $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{mix}}$ also carries ρ . Assume the functions below are bounded and measurable on their declared domains. Define g on $\mathcal{C}_{\text{mix}}$ and c, b on $\mathcal{C}_\pi$ by*

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} [A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{mix}}} g(s) - \inf_{s \in \mathcal{C}_{\text{mix}}} g(s), \quad (20)$$

$$c(s) := \mathbb{E}_{a \sim \pi(\cdot | s)} [\hat{A}(s, a)], \quad b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} [\hat{A}(s, a) - A^\pi(s, a)], \quad (21)$$

$$\Xi_\alpha := \mathbb{E}_{s \sim d_\pi} [(1 - \alpha)c(s) + \alpha b(s)]. \quad (22)$$

Then the surrogate error is exactly

$$\hat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\Xi_\alpha}{1 - \gamma}, \quad (23)$$

and

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\Xi_\alpha}{1 - \gamma} - \frac{\gamma\alpha^2\Delta_g}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (24)$$

Because Ξ_α is signed, a usable one-sided guarantee may replace it by any established $B_\alpha \geq \Xi_\alpha$.

Under exact statewise centering,

$$\mathbb{E}_{a \sim \pi(\cdot | s)} [\hat{A}(s, a)] = 0 \quad \text{for every } s \in \mathcal{C}_\pi, \quad (25)$$

define the uniform candidate-policy bias

$$\varepsilon_{A,\text{cand}} := \sup_{s \in \mathcal{C}_\pi} \left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} \left[\hat{A}(s, a) - A^\pi(s, a) \right] \right|. \quad (26)$$

If $\varepsilon_{A,\text{cand}} < \infty$ and $\varepsilon_{\text{CPI}} := \sup_{s \in \mathcal{C}_{\text{mix}}} |g(s)| < \infty$, then

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\alpha}{1 - \gamma} \varepsilon_{A,\text{cand}} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (27)$$

Exact centering leaves candidate bias scaled by $\alpha/(1 - \gamma)$. Occupancy shift uses Δ_g and the exact denominator $1 - \gamma + \gamma\alpha$. A negative defect credits underestimated advantage; an absolute bound loses that credit (Section 9). For example, $B_\alpha = (1 - \alpha)\mathbb{E}_{d_\pi} |c(s)| + \alpha\mathbb{E}_{d_\pi} |b(s)|$ upper-bounds Ξ_α . Sampled or learned baselines contribute $c(s)$. Proof in Appendix F.7.

The domain and span may depend on α . A common-domain safe-step extension appears in Corollary F.4 (Appendix F.8); Appendix F.9 gives its proof and boundaries. Assumption 7.1 implies $\varepsilon_{A,\text{cand}} \leq \varepsilon_A$, but the signed theorem does not require it.

8 PERSISTENT LATENTS AND DEPLOYMENT

Markovization with a carried latent. Fix the measurable initialization law $\bar{\rho}$, bounded rewards, measurable reward/edit and policy kernels, value head, and shared recurrent map on $\bar{s} = (x, y, z, h)$. Deployment uses F_n for both policies' inputs and carries its output only after a nonterminal transition; every terminal cause reaches the common absorber. For any finite comparison depth $q \geq 0$, the endpoint starts from the same stored z :

$$\tilde{U}_q(\bar{s}) := V_\psi(\tilde{f}_\theta^q(z, y, x), x, y), \quad \tilde{U}_q(s_{\text{abs}}) := -C_{\text{max}}. \quad (28)$$

$$\bar{\pi}(a | \bar{s}) := \pi_\phi(a | y, F_n(x, y, z), x), \quad \bar{\pi}_{\text{cand}}(a | \bar{s}) := \pi_{\text{cand}}(a | y, F_n(x, y, z), x), \quad (29)$$

and let $\bar{T}_K^\pi$ denote the K -step Bellman operator of this augmented MDP, whose transitions use the deployed F_n .

Changing q leaves F_n , both policies in Eq. (29), and the transition kernel fixed. Appendix C.1 and Figure 1 give the construction.

Corollary 8.1 (Persistent-state evaluation and improvement). *Let $\gamma \in [0, 1)$ and $K \geq 1$. Let $\bar{\mathcal{C}}_\pi$ be a measurable $\bar{\pi}$ -invariant domain carrying $\bar{\rho}$, and let $\bar{\mathcal{C}}$ be a measurable augmented advantage-evaluation domain as in Definition A.3, containing $\bar{\mathcal{C}}_\pi$ and the common absorber. Assume the preceding frozen-MDP and measurability conditions, bounded K -step expected rewards, and $\tilde{U}_n \in B_b(\bar{\mathcal{C}})$ with the common scalar boundary. Use the ordinary pointwise norm on $\bar{\mathcal{C}}$ and define*

$$\bar{\varepsilon}_{\text{res},n} := \|\tilde{U}_n - \bar{T}_K^\pi \tilde{U}_n\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (30)$$

The Bellman operator has a unique $V^{\bar{\pi}} \in B_b(\bar{\mathcal{C}})$, and

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{\varepsilon}_{\text{res},n}}{1 - \gamma^K}. \quad (31)$$

If $\hat{A}_{\tilde{U}_n}$ uses the exact transition expectation and exact baseline and is therefore centered pointwise under $\bar{\pi}$ on the full augmented state, then Assumption 7.1 holds for every $\bar{s} \in \bar{\mathcal{C}}_\pi$ and for $\bar{\pi}$ - and $\bar{\pi}_{\text{cand}}$ -almost every action, with the valid choice

$$\bar{\varepsilon}_A := \frac{2\gamma \bar{\varepsilon}_{\text{res},n}}{1 - \gamma^K}. \quad (32)$$

If, additionally, the analyzed policy is the pointwise exact mixture $\bar{\pi}_{\text{new}} = (1 - \alpha)\bar{\pi} + \alpha\bar{\pi}_{\text{cand}}$ with no parameter interpolation or distillation, let $\bar{\mathcal{C}}_{\text{mix}}$ be a measurable augmented domain carrying $\bar{\rho}$, containing $\bar{\mathcal{C}}_\pi$, and invariant under both $\bar{\pi}$ and $\bar{\pi}_{\text{new}}$. Define

$$\bar{\varepsilon}_{\text{CPI}} := \sup_{\bar{s} \in \bar{\mathcal{C}}_{\text{mix}}} \left| \mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot | \bar{s})} A^{\bar{\pi}}(\bar{s}, a) \right|, \quad (33)$$

$$\bar{\tau}_\infty := \sup_{\bar{s} \in \bar{\mathcal{C}}_\pi} \text{TV}(\bar{\pi}_{\text{cand}}(\cdot | \bar{s}), \bar{\pi}(\cdot | \bar{s})), \quad (34)$$

where $\bar{\mathcal{C}}_\pi \subseteq \bar{\mathcal{C}} \cap \bar{\mathcal{C}}_{\text{mix}}$ is the common full-measure current-policy domain. With $d_{\bar{\pi}} = (1 - \gamma) \sum_{t \geq 0} \gamma^t \bar{\rho} P_{\bar{\pi}}^t$ and $\bar{A} = \bar{A}_{\bar{\pi}}$, define the full-state surrogate

$$\hat{L}_{\bar{\pi}}(\bar{\pi}') := \eta(\bar{\pi}) + \frac{1}{1 - \gamma} \mathbb{E}_{\bar{s} \sim d_{\bar{\pi}}, a \sim \bar{\pi}'(\cdot|\bar{s})} [\hat{A}(\bar{s}, a)]. \quad (35)$$

Theorem 7.3 gives

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_\infty}{1 - \gamma} \frac{\bar{\varepsilon}_{\text{res},n}}{1 - \gamma^K} - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (36)$$

For the target-network and finite-reference forms, apply Corollary D.1 with barred quantities: use a finite target depth $q \geq 0$, $\tilde{U}_q, \tilde{V} \in B_b(\bar{\mathcal{C}})$ with the common boundary, a target trajectory under the frozen augmented policy, and exact transition expectation and baseline conditional on the full augmented state. All corresponding domain and exact-mixture hypotheses remain in force.

For finite $0 \leq n < m$ and $\tilde{U}_n, \tilde{U}_m \in B_b(\bar{\mathcal{C}})$ with the common boundary, the same-MDP finite-reference bound is

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_\infty \leq \|\tilde{U}_n - \tilde{U}_m\|_\infty + \frac{\|\tilde{U}_m - \bar{T}_K^{\bar{\pi}} \tilde{U}_m\|_\infty}{1 - \gamma^K}. \quad (37)$$

Proof in Appendix F.10.

Approximating the mixture at deployment. Distillation or parameter/logit/hidden-state interpolation generally gives a different law $\bar{\pi}$. On a domain invariant under both policies, let its uniform TV discrepancy from π_α be at most δ_{dep} , and let Δ_r be the common reward-interval width. Under the standard-Borel and measurable-kernel assumptions of Theorem E.1,

$$|\eta(\bar{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \quad (38)$$

The first-mismatch penalty vanishes for exact realization. Either-direction uniform KL control supplies a TV radius (Corollary E.3). Persistent policies must share the recurrent transition map; changing it requires a different-MDP analysis.

9 AN EXACT EXAMPLE

Exact enumeration gives a nonvacuous value radius and signed CPI bound, but a vacuous absolute specialization. Nothing is sampled or learned.

Consider two nonabsorbing states and a shared absorber, $\mathcal{S} = \{s_0, s_1, s_{\text{abs}}\}$, actions $\{a, b\}$, initial law $\rho = \delta_{s_0}$, $\gamma = 1/2$, and $K = 1$. The current policy always chooses a and the candidate always chooses b . Deterministic reward-successor pairs are

$$(s_0, a) : (0, s_0), \quad (s_0, b) : (1, s_1), \quad (s_1, a) : (0, s_0), \quad (s_1, b) : (0, s_1),$$

and both absorber actions return $(0, s_{\text{abs}})$. The complete domains are $\mathcal{C}_\pi = \{s_0, s_{\text{abs}}\}$ and $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} = \mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}} = \mathcal{D}_{\text{dep}} = \mathcal{S}$. For the scalar recurrence, set $z^{(0)} = 0$, $T_s(z) = u_s - z/2$, $(u_{s_0}, u_{s_1}) = (1/4, -1/4)$, and $u_{s_{\text{abs}}} = 0$, with $V_\psi(z, s) = z$. Use deployed depth $n = 1$, comparison depth $m = 2$, and $\alpha = 1/2$. Table 1 reports the exact result.

The finite-reference radius $B_{1,2} = 1/2$ is twice the actual error $1/4$; it uses two finite endpoints rather than a recurrent limit. For comparison, the self-bootstrap bound at depth 1 (Corollary D.1) gives $\varepsilon_{\text{res},1}/(1 - \gamma) = (3/8)/(1/2) = 3/4$. The fixed-point specialization Eq. (83) applies with $\mathcal{Z}_{\text{inv}} = [-1/2, 1/2]$, $L_z = 1/2$, $C_z = 1/4$, and $L_V = 1$; it gives $\varepsilon_{\text{res}}^*/(1 - \gamma) + L_V L_z^n C_z/(1 - L_z) = 1/2 + 1/4 = 3/4$. Deeper references are not monotone: $m = 3$ gives $5/8$ and $m = 4$ gives $9/16$.

The policy calculation illustrates why signs and spans matter. The estimator understates candidate advantage: $b(s_0) = -1/4$ and $\Xi_{1/2} = -1/8$. Correcting the surrogate $3/4$ therefore adds $1/4$. The exact-span occupancy penalty is $1/3$, so

$$\eta(\pi_{1/2}) \geq \frac{3}{4} + \frac{1}{4} - \frac{1}{3} = \frac{2}{3}, \quad \eta(\pi_{1/2}) = \frac{3}{4}.$$

Table 1: Exact two-nonabsorbing-state calculation. State vectors are ordered $(s_0, s_1, s_{\text{abs}})$. No quantity is sampled or learned.

Quantity	Exact value
$V^\pi; U_1; U_2$	$(0, 0, 0); (1/4, -1/4, 0); (1/8, -1/8, 0)$
$\ U_1 - U_2\ _\infty; \ U_2 - \mathcal{T}_1^\pi U_2\ _\infty$	$1/8; 3/16$
Finite-reference upper bound; actual $\ U_1 - V^\pi\ _\infty$	$1/2; 1/4$
$d_\pi; d_{\pi_{1/2}}; \pi_{1/2}(a \mid s), \pi_{1/2}(b \mid s)$	$(1, 0, 0); (3/4, 1/4, 0); (1/2, 1/2)$ at every state
$g; \Delta_g; c; b$	$(1, 0, 0); 1; (0, 0, 0); (-1/4, -1/4, 0)$
$\hat{L}_\pi(\pi_{1/2}); \Xi_{1/2}$	$3/4; -1/8$
Signed CPI lower bound; actual $\eta(\pi_{1/2})$	$2/3; 3/4$

The absolute uniform specialization instead subtracts $1/4$ for evaluation error and replaces the span $\Delta_g = 1$ by $2\varepsilon_{\text{CPI}} = 2$, doubling the occupancy penalty:

$$\frac{3}{4} - \frac{1}{4} - \frac{2}{3} = -\frac{1}{6}.$$

Both bounds are valid, but only the signed-span version certifies improvement over $\eta(\pi) = 0$. Appendix B derives every quantity; `verify_finite_mdp.py` reproduces the rational calculation.

10 DISCUSSION, LIMITATIONS, AND CONCLUSION

A finite reference separates endpoint discrepancy from self-bootstrap residual and supplies a value radius for one-proposal CPI. Target lag and deployment mismatch require separate corrections; persistent state requires the clock and a shared frozen map. Recurrent convergence is unnecessary.

These conditional guarantees concern one fixed-base proposal; training convergence and learned-task improvement remain open. The exact MDP checks nonvacuity. Finite-batch residuals, centering checks, and deployment measurements are diagnostics and do not establish the uniform premises of the bounds. Ethics, reproducibility, and AI-use statements follow.

AI USE STATEMENT

In this work, we used generative AI tools for [required-disclosure tasks used: e.g., assisting the writing of proofs, formulating or refining mathematical claims, interpreting the exact example]. We have not used generative AI tools for [required-disclosure tasks not used], and [remaining required tasks] are not applicable to this work. Additionally, we used generative AI tools for literature search, editing the manuscript for readability, LaTeX reorganization, figure editing, and preparing cross-reference and exact-arithmetic validation checks. We have reviewed all AI-assisted work: [verification the authors performed, e.g., every proof was checked line by line by the authors and the exact example is reproduced by a standard-library script]. We take responsibility for the final content of this work.

REPRODUCIBILITY STATEMENT

The assumptions are stated with each result, and proofs appear in Appendices F and G.3. Appendix B gives the full rational calculation for Table 1; the standard-library script `verify_finite_mdp.py` reproduces its quantities. Appendix H specifies the implementation semantics.

ETHICS AND BROADER IMPACT

Ethics statement. The paper studies a mathematical model of checker-trained Tiny Recursive Models on synthetic plan-editing tasks. We use no personal data, no human subjects, no crowdsourcing, and no deployed decision system. The main ethical risk is rhetorical: conditional bounds can be overstated as evidence about general reasoning or high-stakes automation.

Broader impact. The methodological contribution is to separate internal evaluator depth from outer policy improvement, which may help analyses of verifier-feedback systems with bounded internal compute. We make no deployment recommendation and keep the claims conditional on the displayed assumptions.

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Table 2: Symbols used in statements

Symbol	Meaning
$h, \bar{s} = (x, y, z, h)$	Remaining edit budget / clock-complete persistent state
$\mathcal{D}^\circ$	Declared nonabsorbing domain for a contraction invocation
$\mathcal{C}, \mathcal{C}^\circ$	Value domain / nonabsorbing latent domain
P_{rec}	Declared recurrent projection operator: Π_R for $R > 0$, or $\text{Id}_{\mathcal{Z}}$
$\mathcal{Z}_{\text{bd}}, V_{\text{max}}$	Declared bounded-head latent set / its uniform finite bound
$z^{(j)}(s), s \in \mathcal{C}^\circ$	Latent after finite depth j
U_j, U_*	Finite-depth value / optional fixed-point value, with scalar boundary
$B_{n,m}$	Endpoint discrepancy plus depth- m residual radius
$\varepsilon_{\text{res},n}$	Episodic finite-depth Bellman residual
$\varepsilon_{\text{targ},q}$	Target-network conditional population residual at endpoint depth q
$R_{\text{targ},q}^P, R_{\text{targ},q}^{\text{sup}}$	Propagated-lag target bound / ordinary sup-lag fallback
$d_{\rho}^{\pi,K}$	Discounted occupancy of the fixed K -step current-policy skeleton
$C_{\rho,\mu}^{\pi,K}, C_{\rho,\mu}^{(0)}$	Fixed-kernel occupancy / initial-law density-ratio bounds for the weighted- L_p restatement
$\bar{\varepsilon}_{\text{res},n}$	Persistent augmented-state finite-depth residual
$\varepsilon_A, \bar{\varepsilon}_A$	Episodic / augmented advantage error bounds
$\varepsilon_{\text{res}}^*$	Architectural residual at U_*
g, Δ_g	Candidate expected advantage / span on the declared mixture or pair domain
$\hat{g}, \bar{\beta}_d, M_d$	Estimated candidate slope / one-sided signed-defect bound / certified slope
$\mathcal{C}_{\text{pair}}$	Common π/π_{cand} -invariant domain for step intervals
$c, b(s), \Xi_\alpha$	Signed centering and candidate estimator defects
$\tau(s), \delta_{\text{dep}}$	Policy overlap / deployment TV radius
ε_{CPI}	Uniform scalar bound on $ g $

A NOTATION, DOMAINS, AND ABSORBING NORMALIZATION

The definitions below separate value evaluation, one-deviation advantage estimation, repeated mixture decisions, and deployment. Table 2 lists the surviving notation.

Assumption A.1 (Bounded rewards). Potential-shaped rewards are uniformly bounded. For some $R_{\text{max}} < \infty$,

$$|r(s, a, s')| \leq R_{\text{max}} \quad \text{for every transition.}$$

For $R > 0$, define the Euclidean projection

$$\begin{aligned} \Pi_R(z) &= \arg \min_{u: \|u\|_2 \leq R} \|u - z\|_2 \\ &= \begin{cases} z, & \|z\|_2 \leq R, \\ R \frac{z}{\|z\|_2}, & \|z\|_2 > R. \end{cases} \end{aligned} \quad (39)$$

The projection Π_R is nonexpansive and leaves B_R invariant. It does not by itself give contraction. The first-increment constant used by the contraction specialization is

$$C_z := \sup_{s \in \mathcal{C}^\circ} \|\tilde{f}_\theta(z_{\text{init}}(x(s), y(s)), y(s), x(s)) - z_{\text{init}}(x(s), y(s))\| < \infty. \quad (40)$$

A.1 EVALUATION AND POLICY DOMAINS

Definition A.2 (π -invariant Bellman domain). A measurable set $\mathcal{C}_\pi$ is a Bellman/value domain for π if $\rho(\mathcal{C}_\pi) = 1$ whenever the initial law is used and

$$P(\mathcal{C}_\pi \mid s, a) = 1 \quad \text{for } \pi(da \mid s)\text{-almost every } a \text{ and every } s \in \mathcal{C}_\pi.$$

Equivalently, the policy-induced kernel leaves $\mathcal{C}_\pi$ invariant. Theorem 5.1 needs only such a domain and does not need a candidate policy.

Table 3: Domain and theorem map. Every domain carries the stated initial law. The superscript $\circ$ removes the absorber only for latent-path, fixed-point, and head-Lipschitz assumptions; value norms include the absorber.

Domain	Required invariance	Paths included	α ?	Used by
$\mathcal{C}_\pi$	P_π	Repeated current-policy decisions	No	Bellman values, d_π , centering
$\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$	P_π after the first action	One current or candidate deviation, then π	No	Finite-reference and value-to-advantage bounds
$\mathcal{C}_{\text{mix}}$	P_π and P_{π_α}	Repeated exact-mixture decisions	May	Theorem 7.3
$\mathcal{C}_{\text{pair}}$	P_π and $P_{\pi_{\text{cand}}}$	Repeated decisions for every exact mixture weight	No	Step-interval corollary
$\mathcal{D}_{\text{dep}}$	P_{π_α} and $P_{\tilde{\pi}}$	Repeated reference-mixture and deployed-policy decisions	Through the selected π_α	Theorem E.1
Barred counterparts	The corresponding augmented kernels with one shared F_n	The same closures on (x, y, z, h)	As above	Persistent-state results

Definition A.3 (Advantage-evaluation domain). Fix current and candidate stochastic kernels π and π_{cand} . A measurable set $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ contains a π -invariant domain $\mathcal{C}_\pi$ and the common absorber and satisfies

1. for every $s \in \mathcal{C}_\pi$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ - and $\pi_{\text{cand}}(da \mid s)$ -almost every action; and
2. for every $s \in \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ -almost every action.

The domain contains current-policy evaluation states, one current or candidate action, and every subsequent current-policy continuation. We write $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ for the ordinary supremum on this set, including the scalar absorbing boundary.

Definition A.4 (Mixture-occupancy and deployment domains). For a fixed α , a measurable $\mathcal{C}_{\text{mix}}$ is a mixture-occupancy domain if $\rho(\mathcal{C}_{\text{mix}}) = 1$ and it is invariant under both π and $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$. It therefore supports the state laws at every time under the current policy and under repeated mixture decisions, not merely a one-deviation path. A deployment domain $\mathcal{D}_{\text{dep}}$ satisfies $\rho(\mathcal{D}_{\text{dep}}) = 1$ and is recursively invariant under both π_α and $\tilde{\pi}$. The CPI span theorem uses $\mathcal{C}_{\text{mix}}$; the deployment theorem uses $\mathcal{D}_{\text{dep}}$.

For a certificate simultaneous over more than one mixture weight, a measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ satisfies

$$\rho(\mathcal{C}_{\text{pair}}) = 1, \quad P_\pi(\mathcal{C}_{\text{pair}} \mid s) = P_{\pi_{\text{cand}}}(\mathcal{C}_{\text{pair}} \mid s) = 1 \quad \text{for every } s \in \mathcal{C}_{\text{pair}}. \quad (41)$$

By linearity of policy-induced kernels, every exact mixture π_α also leaves $\mathcal{C}_{\text{pair}}$ invariant. Thus $\mathcal{C}_{\text{pair}}$ is a valid single choice of $\mathcal{C}_{\text{mix}}$ for every $\alpha \in [0, 1]$.

In finite or countable spaces these domains can be constructed with the usual positive-probability reachable sets and action supports. On arbitrary measurable action spaces we use the policy-almost-every-action clauses above; no topological action support is presumed.

Why boundedness matters. On the countable zero-reward chain $s_i \mapsto s_{i+1}$ with one action and $\gamma \in (0, 1)$, add an unreachable zero-reward absorber with $U(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = 0$. The unbounded function $U(s_i) = \gamma^{-i}$ has zero K -step residual for every $K \geq 1$, while its sup-norm error from $V^\pi = 0$ is infinite. This is why the residual theorem works on bounded functions, not arbitrary pointwise solutions.

A.2 SHAPING CONVENTION AND ABSORBING-STATE NORMALIZATION

We use potential-based shaping with potential $\Phi(s) = c(x, y)$ for episodic $s = (x, y, h)$ and persistent $\bar{s} = (x, y, z, h)$ (the potential ignores the clock and latent), and $\Phi(s_{\text{abs}}) = C_{\text{max}}$ on their shared absorbing atom. We define

$$r(s, a, s') = r_0(s, a, s') + \gamma \Phi(s') - \Phi(s), \quad (42)$$

applying shaping on *all* transitions, including the absorbing self-loop.

For the absorbing self-loop, taking $r_0(s_{\text{abs}}, a, s_{\text{abs}}) = 0$ gives

$$r(s_{\text{abs}}, a, s_{\text{abs}}) = \gamma\Phi(s_{\text{abs}}) - \Phi(s_{\text{abs}}) = (\gamma - 1)C_{\text{max}}.$$

Thus every policy satisfies the fixed shaped boundary value

$$V^\pi(s_{\text{abs}}) = \sum_{t=0}^{\infty} \gamma^t (\gamma - 1) C_{\text{max}} = -C_{\text{max}}.$$

If one prefers the standard normalization $V(s_{\text{abs}}) = 0$, use the shifted potential $\tilde{\Phi}(s) := \Phi(s) - C_{\text{max}}$. This adds the constant reward $(1 - \gamma)C_{\text{max}}$ to every transition and shifts all returns by $+C_{\text{max}}$ independently of the policy. Accordingly, performance differences and optimal policies are unchanged.

B EXACT FINITE-MDP DERIVATION

This section derives Table 1. It uses exact rational arithmetic and no data, learned parameter, or random sample. The state and action definitions are

State	Action	(r, S')
s_0	a	$(0, s_0)$
s_0	b	$(1, s_1)$
s_1	a	$(0, s_0)$
s_1	b	$(0, s_1)$
s_{abs}	a or b	$(0, s_{\text{abs}})$

The current policy is $\pi(a | s) = 1$, the candidate is $\pi_{\text{cand}}(b | s) = 1$, and $\rho = \delta_{s_0}$. These definitions also apply at the absorber, where both actions induce the same zero-reward self-loop. Thus

$$\mathcal{C}_\pi = \{s_0, s_{\text{abs}}\}, \quad \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} = \mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}} = \mathcal{D}_{\text{dep}} = \mathcal{S}.$$

The first set is π -invariant and carries ρ . One candidate deviation from s_0 reaches s_1 , after which π returns to s_0 . Repeated mixture decisions remain in $\mathcal{S}$, so this is the complete relevant closure.

Finite-reference value calculation. Set $\gamma = 1/2$, $K = 1$, $n = 1$, and $m = 2$. On the scalar latent space, use the common initialization $z^{(0)}(s) = 0$, head $V_\psi(z, s) = z$, and

$$T_s(z) = u_s - \frac{1}{2}z, \quad (u_{s_0}, u_{s_1}, u_{s_{\text{abs}}}) = \left(\frac{1}{4}, -\frac{1}{4}, 0\right).$$

The absorber therefore has the common depth-independent boundary zero. The finite endpoints are

$$U_1 = \left(\frac{1}{4}, -\frac{1}{4}, 0\right), \quad U_2 = \left(\frac{1}{8}, -\frac{1}{8}, 0\right).$$

Under π , all nonabsorbing rewards are zero and both nonabsorbing states move to s_0 , so $V^\pi = (0, 0, 0)$. Moreover,

$$\mathcal{T}_1^\pi U_2 = \left(\frac{1}{16}, \frac{1}{16}, 0\right), \quad U_2 - \mathcal{T}_1^\pi U_2 = \left(\frac{1}{16}, -\frac{3}{16}, 0\right).$$

Consequently,

$$\begin{aligned} \|U_1 - U_2\|_\infty &= \frac{1}{8}, & \|U_2 - \mathcal{T}_1^\pi U_2\|_\infty &= \frac{3}{16}, \\ \|U_1 - V^\pi\|_\infty &\leq \frac{1}{8} + \frac{3/16}{1 - 1/2} = \frac{1}{2}, & \|U_1 - V^\pi\|_\infty &= \frac{1}{4}. \end{aligned}$$

The certificate gives the finite error radius $1/2$ without a recurrent fixed-point argument. It is not tight: the actual error is smaller by a factor of two.

Exact mixture and occupancies. Take $\alpha = 1/2$. At every state, $\pi_{1/2}(a | s) = \pi_{1/2}(b | s) = 1/2$. On the two nonabsorbing states,

$$P_{\pi_{1/2}} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \quad r_{\pi_{1/2}} = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}.$$

The absorber is a separate zero-reward self-loop. Solving $d = (1 - \gamma)\rho + \gamma dP$ gives

$$d_{\pi} = (1, 0, 0), \quad d_{\pi_{1/2}} = \left(\frac{3}{4}, \frac{1}{4}, 0\right).$$

Solving the mixture Bellman equations gives $V^{\pi_{1/2}} = (3/4, 1/4, 0)$ and hence $\eta(\pi_{1/2}) = 3/4$.

Advantage, centering, and CPI lower bound. Because $V^{\pi} = 0$, the candidate expected advantage is

$$g = (1, 0, 0), \quad \Delta_g = 1.$$

For the value-derived one-step estimator based on U_1 , the exact action values on (s_0, s_1) are

$$\hat{Q}_{U_1} = \begin{pmatrix} 1/8 & 7/8 \\ 1/8 & -1/8 \end{pmatrix},$$

with columns (a, b) . Exact centering under the deterministic current policy subtracts the first column. Including the absorber gives

$$\hat{A}(a) = (0, 0, 0), \quad \hat{A}(b) = \left(\frac{3}{4}, -\frac{1}{4}, 0\right).$$

Thus the centering defect and candidate defect from Eq. (21) are

$$c = (0, 0, 0), \quad b = \left(-\frac{1}{4}, -\frac{1}{4}, 0\right), \quad \Xi_{1/2} = -\frac{1}{8}.$$

The estimated surrogate is

$$\hat{L}_{\pi}(\pi_{1/2}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi}, a \sim \pi_{1/2}} \hat{A}(s, a) = 2 \left(\frac{1}{2} \cdot \frac{3}{4}\right) = \frac{3}{4}.$$

The exact span penalty is

$$\frac{\gamma \alpha^2 \Delta_g}{(1 - \gamma)(1 - \gamma + \gamma \alpha)} = \frac{1}{3}.$$

Equation (24) therefore yields

$$\eta(\pi_{1/2}) \geq \frac{3}{4} - \frac{-1/8}{1/2} - \frac{1}{3} = \frac{2}{3},$$

while the exact return is $3/4$. In contrast, the absolute uniform specialization uses $\varepsilon_{A, \text{cand}} = 1/4$ and $\varepsilon_{\text{CPI}} = 1$, producing

$$\frac{3}{4} - \frac{1}{4} - \frac{2}{3} = -\frac{1}{6}.$$

This valid but vacuous bound shows why the signed defect and the advantage span matter. Run `python3 UPI_TRM_ICLR/verify_finite_mdp.py` from the repository root to reproduce the calculation.

C PERSISTENT LATENTS ON THE AUGMENTED MDP

C.1 AUGMENTED STATE AND ENDPOINT CONSTRUCTION

Corollary 8.1 uses the following augmented MDP construction, which permits a continuous carried latent $z \in \mathbb{R}^d$.

Fix n , (θ, ψ) , the policy parameters, and all environment kernels. Let $h \in \{1, 2, \dots, H\}$ be the number of edits remaining before the current action. Define

$$F_n(x, y, z) := T_{x,y}^{\circ n}(z) = \tilde{f}_{\theta}^{\circ n}(z, y, x), \quad \bar{s} = (x, y, z, h). \quad (43)$$

For a nonabsorbing state, first compute F_n and sample an action. If the environment transition terminates for any reason (including a solution or terminal STOP), or if $h = 1$, the successor is the shared atom s_{abs} ; budget exhaustion is terminal. Otherwise the clock-complete successor is

$$\bar{s}' = (x, \text{edit}(y, a; x), F_n(x, y, z), h - 1).$$

The state stores the pre-unroll carried latent z , the policy uses the post-unroll latent $F_n(x, y, z)$, and a nonterminal successor carries that post-unroll latent. The current and candidate action kernels share this same frozen F_n and hence the same augmented MDP, as in Eq. (29). Changing the recurrent map between these two policies would change the transition kernel and is outside ordinary CPI.

Remark C.1 (Why the recurrent map is shared). A small Euclidean discrepancy between deterministic recurrent maps does not imply a small policy or augmented-kernel discrepancy. For example, take $F_\varepsilon(z) = -\varepsilon/2$ and $G_\varepsilon(z) = \varepsilon/2$ and a fixed threshold policy head that chooses disjoint actions below and above zero. Then $\sup_z \|F_\varepsilon(z) - G_\varepsilon(z)\| = \varepsilon$, but the action laws are disjoint and a fixed MDP can have an order- $1/(1 - \gamma)$ return gap. Moreover, $\text{TV}(\delta_{F(z)}, \delta_{G(z)}) = 1$ whenever $F(z) \neq G(z)$. A different-map theorem therefore needs explicit reward and transition-kernel discrepancies and sufficient regularity to propagate them. Map distance alone is insufficient.

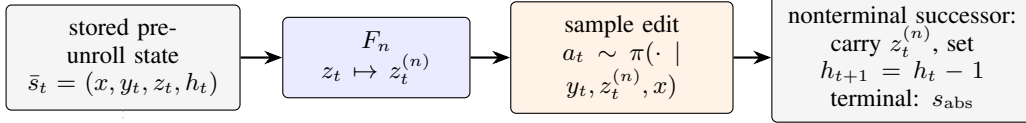
The initial law is

$$\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh) \delta_{z_{\text{init}}(x, y)}(dz), \quad (44)$$

or, for stochastic initialization, $\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh) \nu_0(dz \mid x, y, h)$ with any recorded measurable initialization kernel. Ignoring h is an architectural specialization, not a Bellman requirement.

The measurable domains and bounded endpoints are those of Corollary 8.1. Its Bellman operator $\bar{\mathcal{T}}_K^\pi$ uses the deployed F_n at every environment transition; comparison depth q changes only the endpoint in Eq. (28). The direct residual and surrogate are Eqs. (30) and (35), with discounted return measured from $\bar{\rho}$.

Deployment: act and advance the MDP



Comparison: read another endpoint only

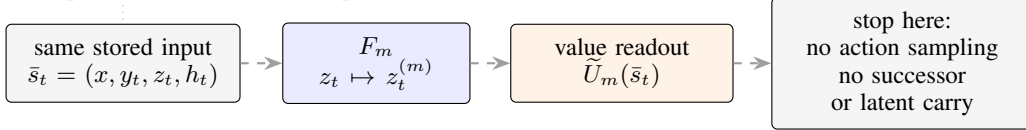


Figure 1: A deeper comparison is not a deeper deployment. Both lanes start from the same stored pre-unroll state. The upper lane uses F_n to choose an action and carries its output only after a nonterminal transition. The lower lane reads another value endpoint and stops: it neither samples an action nor changes the successor. Changing m leaves the deployed MDP and trajectory unchanged.

Comparison with a memoryless fixed-point reference would require a drift assumption that this paper does not make.

D RESIDUAL-TO-CPI CERTIFICATES

Corollary D.1 (Self-bootstrap and target-network residual certificates). *Use the measurable Bellman domain of Theorem 5.1 with $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. Take $U_n \in B_b(\mathcal{C})$ and use the common scalar boundary $U_n(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = b$. Define*

$$\varepsilon_{\text{res}, n} := \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (45)$$

Then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}, n}}{1 - \gamma^K}. \quad (46)$$

If $\hat{A} = \hat{A}_{U_n}$ is the exact one-step centered estimator of Eq. (2), with its transition expectation and baseline sum under π evaluated exactly, then Assumption 7.1 holds with

$$\varepsilon_A := \frac{2\gamma \varepsilon_{\text{res},n}}{1 - \gamma^K}. \quad (47)$$

Under Theorem 7.3,

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \frac{\varepsilon_{\text{res},n}}{1 - \gamma^K} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (48)$$

For a finite depth $q \geq 0$, instead suppose Proposition 6.1 holds on the same domain with $\mathcal{E} = \mathcal{C}$. Then

$$\|U_q - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{R_{\text{targ},q}^P}{1 - \gamma^K}. \quad (49)$$

For finite depths $0 \leq n < m$, if $U_n, U_m \in B_b(\mathcal{C})$ use the same scalar boundary and the depth- m target-network quantities are established, then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ},m}^P}{1 - \gamma^K}. \quad (50)$$

For the depth- q target statement, take $\hat{A} = \hat{A}_{U_q}$ from Eq. (2), with its transition expectation and current-policy baseline evaluated exactly, and define $\varepsilon_{A,\text{cand}}$ by Eq. (26). Then

$$\varepsilon_{A,\text{cand}} \leq 2\gamma\tau_\infty \frac{R_{\text{targ},q}^P}{1 - \gamma^K}. \quad (51)$$

Under the exact-mixture and scalar-span hypotheses of Theorem 7.3,

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{(1 - \gamma)(1 - \gamma^K)} R_{\text{targ},q}^P - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}, \quad (52)$$

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \left(\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ},m}^P}{1 - \gamma^K} \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (53)$$

The second display uses the same exact value-derived construction at depth n . Every target expression retains the propagated-lag quantity $R_{\text{targ},q}^P$ before the ordinary sup-lag fallback $R_{\text{targ},q}^P \leq R_{\text{targ},q}^{\text{sup}}$.

Proof in Appendix F.15. The common-domain safe-step result is Corollary F.4.

E DEPLOYMENT MISMATCH

Theorem E.1 (First-mismatch deployment perturbation). Fix $0 \leq \gamma < 1$ and $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ for $\alpha \in [0, 1]$. Let $\tilde{\pi}$ be a deployed stochastic kernel in the same fixed MDP as π_α , with the same initial law ρ . Assume the state and action spaces are standard Borel, and let $K(dr, ds' | s, a)$ be the measurable joint reward-transition kernel.

Let $\mathcal{D}_{\text{dep}}$ be measurable with $\rho(\mathcal{D}_{\text{dep}}) = 1$, and assume the policy-induced transition kernels of both π_α and $\tilde{\pi}$ leave $\mathcal{D}_{\text{dep}}$ invariant. For every $s \in \mathcal{D}_{\text{dep}}$, assume K assigns rewards in $[r_{\min}, r_{\max}]$ for $\pi_\alpha(da | s)$ - and $\tilde{\pi}(da | s)$ -almost every action. Write $\Delta_r := r_{\max} - r_{\min}$ and suppose

$$\sup_{s \in \mathcal{D}_{\text{dep}}} \text{TV}(\tilde{\pi}(\cdot | s), \pi_\alpha(\cdot | s)) \leq \delta_{\text{dep}}, \quad \delta_{\text{dep}} \in [0, 1]. \quad (54)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma\delta_{\text{dep}})}. \quad (55)$$

The extra denominator factor satisfies $1 - \gamma + \gamma\delta_{\text{dep}} \geq 1 - \gamma$, so the first-mismatch constant improves the generic perturbation constant.

Proof in Appendix F.12.

Remark E.2 (Attainment by a first-mismatch process). The constant in Theorem E.1 is attained by a two-state MDP. Let ℓ be initial and u absorbing. At ℓ , action a_0 gives reward 0 and returns to ℓ , while a_1 gives reward Δ_r and enters u ; every action at u gives reward Δ_r and remains there. Let π_α choose a_0 at ℓ , and let $\tilde{\pi}$ choose a_1 with probability δ_{dep} and a_0 otherwise. Let the policies agree at u . The reward difference at time t is exactly $\Delta_r[1 - (1 - \delta_{\text{dep}})^{t+1}]$, so equality holds in Eq. (55).

Corollary E.3 (Either-direction uniform-KL deployment bound). *Under Theorem E.1, suppose either*

$$\begin{aligned} \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\tilde{\pi}(\cdot | s) \| \pi_\alpha(\cdot | s)) &\leq \kappa, \quad \text{or} \\ \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\pi_\alpha(\cdot | s) \| \tilde{\pi}(\cdot | s)) &\leq \kappa \end{aligned}$$

for a finite κ . Define

$$\bar{\delta}_\kappa := \min \left\{ 1, \sqrt{\frac{\kappa}{2}} \right\}. \quad (56)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \bar{\delta}_\kappa}{(1 - \gamma)(1 - \gamma + \gamma \bar{\delta}_\kappa)}. \quad (57)$$

Proof in Appendix F.13.

Corollary E.4 (Occupancy-aware CPI with deployment discrepancy). *Assume Theorems 7.3 and E.1 hold in one fixed MDP with one initial law. Let $\mathcal{C}_{\text{mix}}$ satisfy Definition A.4, and let $\mathcal{D}_{\text{dep}}$ be measurable, carry the same initial law, and be recursively invariant under both π_α and $\tilde{\pi}$. Assume the standard-Borel, bounded-reward, and uniform-TV conditions of Theorem E.1. For any established one-sided bound $B_\alpha \geq \Xi_\alpha$,*

$$\begin{aligned} \eta(\tilde{\pi}) &\geq \hat{L}_\pi(\pi_\alpha) - \frac{B_\alpha}{1 - \gamma} - \frac{\gamma \alpha^2 \Delta_g}{(1 - \gamma)(1 - \gamma + \gamma \alpha)} \\ &\quad - \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \end{aligned} \quad (58)$$

A distilled, logit-interpolated, hidden-state-interpolated, parameter-interpolated, or otherwise realized policy is covered only when it acts in this same MDP and its TV or KL discrepancy from the exact probability-space mixture is established uniformly on $\mathcal{D}_{\text{dep}}$. For persistent state (x, y, z, h) , all policies must use the same frozen recurrent map that determines the carried-latent transition. Changing that map changes the MDP.

Proof in Appendix F.14.

F PROOFS

F.1 FINITE-REFERENCE VALUE BOUND

Proof of Theorem 5.1. Write $\|\cdot\|_\infty$ for the ordinary sup-norm on $\mathcal{C}$. Invariance and the Markov-kernel measurability property imply $P_\pi^K V \in B_b(\mathcal{C})$ whenever $V \in B_b(\mathcal{C})$; bounded r_K^π therefore makes $\mathcal{T}_K^\pi$ a self-map. For $V, W \in B_b(\mathcal{C})$,

$$\|\mathcal{T}_K^\pi V - \mathcal{T}_K^\pi W\|_\infty \leq \gamma^K \|V - W\|_\infty.$$

The bounded-reward premise makes the ordinary discounted policy value well defined, bounded, and measurable, with the uniformly convergent representation

$$V^\pi = \sum_{t=0}^{\infty} \gamma^t P_\pi^t r^\pi.$$

Splitting this series after K terms gives $V^\pi = \mathcal{T}_K^\pi V^\pi$. Completeness and Banach's theorem then identify this policy value as the unique fixed point of $\mathcal{T}_K^\pi$. Thus

$$\begin{aligned} \|U_m - V^\pi\|_\infty &\leq \|U_m - \mathcal{T}_K^\pi U_m\|_\infty + \|\mathcal{T}_K^\pi U_m - \mathcal{T}_K^\pi V^\pi\|_\infty \\ &\leq \|U_m - \mathcal{T}_K^\pi U_m\|_\infty + \gamma^K \|U_m - V^\pi\|_\infty. \end{aligned}$$

Rearranging and then applying the triangle inequality to $U_n - U_m + U_m - V^\pi$ proves (6). $\square$

With a finite edit budget, a common terminal boundary and uniformly controlled stagewise residuals, the same backward recursion gives a finite geometric sum, including the final partial block; this refinement is not used here.

Noncontractive endpoints. The scalar map $F(z) = -z$ is nonconvergent away from zero, while endpoints whose depths have the same parity agree exactly. The finite-reference bound still requires the reference endpoint’s Bellman residual.

F.2 RECURRENT ENDPOINT PATHS

Corollary F.1 (Recurrent endpoint path bound). *Under Theorem 5.1, take $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ and $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$. On $\mathcal{C}^\circ$, fix a jointly measurable recurrent map $(s, z) \mapsto T(s, z)$, a common measurable initialization into its declared latent domain, and a latent norm, and define $z^{(j)}$ by iterating that map and $U_j(s) = V_\psi(z^{(j)}(s), x(s), y(s))$. At the absorber define only the common scalar boundary $U_j(s_{\text{abs}}) = b$, and assume the two endpoint functions lie in $B_b(\mathcal{C})$. If the measurable scalar head is L_V -Lipschitz in its latent argument uniformly while (x, y) is held fixed, then*

$$\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V \sup_{s \in \mathcal{C}^\circ} \|z^{(n)}(s) - z^{(m)}(s)\| \quad (59)$$

$$\leq L_V \sup_{s \in \mathcal{C}^\circ} \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\| \quad (60)$$

$$\leq L_V \sum_{j=n}^{m-1} \sup_{s \in \mathcal{C}^\circ} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \quad (61)$$

If $\mathcal{C}^\circ$ is empty, the nonnegative latent suprema in this chain are defined to be zero.

Proof of Corollary F.1. For every nonabsorbing s , Lipschitzness of the scalar head and the telescoping latent path give

$$\begin{aligned} |U_n(s) - U_m(s)| &\leq L_V \|z^{(n)}(s) - z^{(m)}(s)\| \\ &\leq L_V \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \end{aligned}$$

The left side is zero at s_{abs} because its value is depth independent. Taking the supremum only over $\mathcal{C}^\circ$ for the latent endpoints proves Eq. (59). The triangle inequality for the telescoping latent path proves Eq. (60); using $\sup_s \sum_j g_j(s) \leq \sum_j \sup_s g_j(s)$ proves Eq. (61). No absorbing latent is defined. $\square$

F.3 VALUE-TO-ADVANTAGE ERROR

Lemma F.2 (Value to one-step advantage error). *Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ be the advantage-evaluation closure from Definition A.3, and let $\delta_V := \|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Then for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a ,*

$$|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V.$$

Proof of Lemma F.2. For any (s, a) ,

$$\hat{Q}_V(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V(s') - V^\pi(s')].$$

If $s \in \mathcal{C}_\pi$, Definition A.3 gives $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} | s, a) = 1$ for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a . Thus $|V(s') - V^\pi(s')| \leq \delta_V$ for transition-almost every s' , and hence $|\hat{Q}_V(s, a) - Q^\pi(s, a)| \leq \gamma \delta_V$.

Similarly,

$$\left| \mathbb{E}_{b \sim \pi(\cdot | s)} \hat{Q}_V(s, b) - \mathbb{E}_{b \sim \pi(\cdot | s)} Q^\pi(s, b) \right| \leq \gamma \delta_V,$$

since $b \sim \pi(\cdot|s)$ and the same almost-sure closure condition applies. Subtracting these two bounds yields $|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma\delta_V$. $\square$

Proof of Corollary 5.2. Theorem 5.1 gives value error at most $B_{n,m}$; Lemma F.2 multiplies this radius by 2γ . $\square$

Remark F.3 (Extensions to K -step and GAE-style advantages). Lemma F.2 is stated for the *one-step* estimator (2), hence the factor 2γ . If instead one defines a K -step action-value estimate

$$\hat{Q}_V^{(K)}(s, a) := \mathbb{E} \left[\sum_{t=0}^{K-1} \gamma^t r_t + \gamma^K V(s_K) \mid s_0 = s, a_0 = a, \pi \right],$$

and the corresponding centered advantage $\hat{A}_V^{(K)}(s, a) := \hat{Q}_V^{(K)}(s, a) - \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V^{(K)}(s, b)]$ (with the baseline computed exactly), then the same argument yields

$$|\hat{A}_V^{(K)}(s, a) - A^\pi(s, a)| \leq 2\gamma^K \delta_V.$$

λ -return or GAE-style estimators lie outside the displayed bridge unless their estimator-specific uniform error is separately established. Such an error may be supplied directly as ε_A in Assumption 7.1; no GAE coefficient is derived here.

F.4 PROOF OF COROLLARY 5.3

Proof of Corollary 5.3. Let $q_e(s, a) = \gamma \mathbb{E}[e(S') \mid s, a]$. Exact centering gives $b(s) = \mathbb{E}_{\pi_{\text{cand}}} q_e - \mathbb{E}_{\pi} q_e$. By Definition A.3, there is a common action set of full $\pi(\cdot|s)$ - and $\pi_{\text{cand}}(\cdot|s)$ -measure on which $q_e(s, a) \in [-\gamma\delta_V, \gamma\delta_V]$. The total-variation expectation inequality on this common essential range proves Eq. (10); taking a supremum or a d_π expectation gives the two corollaries. For a K -step estimator, $q_e^{(K)}(s, a) = \gamma^K \mathbb{E}[e(S_K) \mid s, a]$ has the corresponding essential range $[-\gamma^K\delta_V, \gamma^K\delta_V]$. $\square$

F.5 TARGET-NETWORK RESIDUALS

Proof of Proposition 6.1. Add and subtract the target-network population backup:

$$U_q - \mathcal{T}_K^\pi U_q = [U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})] + \gamma^K P_\pi^K (\bar{V} - U_q).$$

The triangle inequality proves Eq. (16). Define the ordinary sup-lag fallback by

$$R_{\text{targ},q}^{\text{sup}} := \varepsilon_{\text{targ},q} + \gamma^K \|\bar{V} - U_q\|_{\infty, \mathcal{E}}. \quad (62)$$

Invariance makes P_π^K a Markov operator on $B_b(\mathcal{E})$, so nonexpansiveness gives $R_{\text{targ},q}^P \leq R_{\text{targ},q}^{\text{sup}}$. $\square$

F.6 PROOF OF LEMMA 7.2

Proof of Lemma 7.2. Realize each mixture decision with an independent Bernoulli selector. The event that decisions $0, \dots, t-1$ all select the current-policy component has probability $(1-\alpha)^t$ and, conditional on that event, produces the current-policy state law before decision t . Therefore

$$\text{TV}(\rho P_{\pi_\alpha}^t, \rho P_\pi^t) \leq 1 - (1-\alpha)^t.$$

Convexity of total variation and the geometric identity give

$$\begin{aligned} \text{TV}(d_{\pi_\alpha}, d_\pi) &\leq (1-\gamma) \sum_{t=0}^{\infty} \gamma^t [1 - (1-\alpha)^t] \\ &= \frac{\gamma\alpha}{1-\gamma+\gamma\alpha}. \end{aligned}$$

The exponent is t , not $t+1$, because occupancy is measured before decision t . The formula gives zero at $\alpha = 0$ or $\gamma = 0$ and γ at $\alpha = 1$, and improves the generic bound because $1-\gamma+\gamma\alpha \geq 1-\gamma$. In the countable case this gives $2\gamma\alpha/(1-\gamma+\gamma\alpha) \leq 2\gamma\alpha/(1-\gamma)$. $\square$

F.7 CPI WITH SIGNED EVALUATION ERROR

Proof of Theorem 7.3. Linearity of the exact mixture and $\mathbb{E}_{a \sim \pi(\cdot|s)} A^\pi(s, a) = 0$ give

$$\begin{aligned} \widehat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\pi} [(1-\alpha)c(s) + \alpha b(s)] \\ &= \frac{\Xi_\alpha}{1-\gamma}, \end{aligned}$$

which proves Eq. (23).

The performance-difference identity also gives

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\alpha}{1-\gamma} (\mathbb{E}_{d_{\pi_\alpha}} g - \mathbb{E}_{d_\pi} g).$$

For any bounded measurable g and probability measures p, q , $|\mathbb{E}_p g - \mathbb{E}_q g| \leq \text{span}(g) \text{TV}(p, q)$. Applying this inequality on $\mathcal{C}_{\text{mix}}$ and then Lemma 7.2 yields

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) \geq -\frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}.$$

Combining the last display with the exact surrogate identity proves Eq. (24). Any $B_\alpha \geq \Xi_\alpha$ may replace the signed term. Under exact centering, $c = 0$ and $\Xi_\alpha = \alpha \mathbb{E}_{d_\pi} b \leq \alpha \varepsilon_{A, \text{cand}}$. Finally, $\Delta_g \leq 2 \sup_s |g(s)| = 2\varepsilon_{\text{CPI}}$, proving Eq. (27). An occupancy-averaged absolute choice is

$$B_\alpha = (1-\alpha)\mathbb{E}_{d_\pi} |c(s)| + \alpha \mathbb{E}_{d_\pi} |b(s)|, \quad (63)$$

which upper-bounds Ξ_α ; substituting it in Eq. (24) gives the absolute-defect specialization. $\square$

F.8 SAFE MIXTURE STEPS

Corollary F.4 (Span-based safe mixture step). *In the setting of Theorem 7.3, fix one measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ from Definition A.4, with $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{pair}}$. Use one fixed estimator $\widehat{A}$ satisfying exact statewise centering on $\mathcal{C}_\pi$ for every $\alpha \in [0, 1]$, and define its fixed candidate-defect function*

$$b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\widehat{A}(s, a) - A^\pi(s, a)]. \quad (64)$$

Neither the estimator nor this defect object depends on α . Define, once on $\mathcal{C}_{\text{pair}}$,

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{pair}}} g(s) - \inf_{s \in \mathcal{C}_{\text{pair}}} g(s), \quad (65)$$

and assume this function is bounded and measurable there. Let $\overline{\beta}_d$ be any finite, α -independent, established one-sided bound satisfying

$$\overline{\beta}_d \geq \mathbb{E}_{s \sim d_\pi} b(s). \quad (66)$$

Define

$$\widehat{g} := \mathbb{E}_{s \sim d_\pi, a \sim \pi_{\text{cand}}(\cdot|s)} [\widehat{A}(s, a)], \quad M_d := \widehat{g} - \overline{\beta}_d. \quad (67)$$

Then, simultaneously for every $\alpha \in [0, 1]$,

$$\eta(\pi_\alpha) - \eta(\pi) \geq \frac{\alpha}{1-\gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right]. \quad (68)$$

Thus $\eta(\pi_\alpha) \geq \eta(\pi)$ is certified whenever

$$M_d \geq \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha}. \quad (69)$$

Three immediate choices preserve the same fixed defect object:

1. *the exact signed defect, $\overline{\beta}_d = \mathbb{E}_{d_\pi} b(s)$;*
2. *the occupancy-averaged absolute defect, $\overline{\beta}_d = \mathbb{E}_{d_\pi} |b(s)|$;*

3. the uniform specialization already used above, $\bar{\beta}_d = \varepsilon_{A,\text{cand}}$ from Eq. (26).

The third choice recovers the previous uniform statement as an immediate specialization. Appendix F.9 gives the exact positive-step threshold and all boundary cases. Under Corollaries 5.2 and 5.3, with the finite-reference estimator $\hat{A} = \hat{A}_{U_n}$, one may use $\mathbb{E}_{d_\pi}|b(s)| \leq 2\gamma B_{n,m}\mathbb{E}_{d_\pi}\tau(s)$ for the occupancy-averaged specialization, or $\varepsilon_{A,\text{cand}} \leq 2\gamma B_{n,m}\tau_\infty$ for the uniform specialization. The bound $\tau_\infty \leq 1$ recovers the previous uniform fallback.

F.9 SAFE-STEP BOUNDARY CASES

For Corollary F.4, the zero step $\alpha = 0$ is equality. At $\alpha = 1$, the bound certifies the full candidate step exactly when $M_d \geq \gamma\Delta_g$. The remaining nonredundant cases are:

- if $\gamma = 0$ and $M_d \geq 0$, every $\alpha \in [0, 1]$ is certified;
- if $M_d < 0$, no positive step is certified by this bound;
- if $\gamma > 0$, $M_d = 0$, and $\Delta_g > 0$, no positive step is certified;
- if $M_d = \Delta_g = 0$, every step has a zero lower bound;
- if $\gamma > 0$, $M_d > 0$, and $M_d \geq \gamma\Delta_g$, every $\alpha \in [0, 1]$ is certified;
- if $\gamma > 0$ and $0 < M_d < \gamma\Delta_g$, every

$$\alpha \leq \frac{M_d(1 - \gamma)}{\gamma(\Delta_g - M_d)} \quad (70)$$

is certified. The denominator is positive and the threshold lies strictly between zero and one.

Proof of Corollary F.4. The invariance of $\mathcal{C}_{\text{pair}}$ under both endpoint policies implies its invariance under every π_α . Hence Theorem 7.3 applies for each $\alpha \in [0, 1]$ with the same choice $\mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}}$, the same Δ_g , estimator, defect function, and one-sided defect bound. Exact centering and the fixed candidate-defect definition give

$$\hat{g} = \mathbb{E}_{s \sim d_\pi}[g(s) + b(s)]. \quad (71)$$

Consequently,

$$\mathbb{E}_{d_\pi}g = \hat{g} - \mathbb{E}_{d_\pi}b \geq \hat{g} - \bar{\beta}_d = M_d. \quad (72)$$

The performance-difference identity, the span/TV expectation inequality, and Lemma 7.2 now give

$$\eta(\pi_\alpha) - \eta(\pi) = \frac{\alpha}{1 - \gamma} \mathbb{E}_{d_{\pi_\alpha}}g \geq \frac{\alpha}{1 - \gamma} \left[\mathbb{E}_{d_\pi}g - \frac{\gamma\alpha\Delta_g}{1 - \gamma + \gamma\alpha} \right] \geq \frac{\alpha}{1 - \gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1 - \gamma + \gamma\alpha} \right].$$

This proves Eqs. (68) and (69). The exact signed choice meets Eq. (66) with equality, while $\mathbb{E}b \leq \mathbb{E}|b| \leq \varepsilon_{A,\text{cand}}$ proves the other two specializations. For

$$\varphi(\alpha) := \frac{\gamma\alpha\Delta_g}{1 - \gamma + \gamma\alpha}, \quad \varphi'(\alpha) = \frac{\gamma\Delta_g(1 - \gamma)}{(1 - \gamma + \gamma\alpha)^2} \geq 0.$$

Thus $\varphi(0) = 0$ and $\max_{\alpha \in [0,1]} \varphi(\alpha) = \varphi(1) = \gamma\Delta_g$. If $\gamma > 0$ and $0 < M_d < \gamma\Delta_g$, then $M_d < \Delta_g$ because $\gamma < 1$, so $\Delta_g - M_d > 0$ and the threshold is positive. Moreover,

$$\frac{M_d(1 - \gamma)}{\gamma(\Delta_g - M_d)} < 1 \iff M_d < \gamma\Delta_g,$$

which is the branch premise. Rearrangement gives Eq. (70); monotonicity and direct substitution give the listed boundary cases. $\square$

F.10 PERSISTENT-STATE EVALUATION AND IMPROVEMENT

Proof of Corollary 8.1. The clock-complete state, frozen shared map F_n , and specified initial law define one augmented MDP for both action kernels. On its bounded measurable functions, $\bar{\mathcal{T}}_K^\pi$ is a γ^K -contraction in the augmented closure norm. In this proof, $\|\cdot\|_\infty^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}$. Thus

$$\begin{aligned} \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}} &\leq \|\tilde{U}_n - \bar{\mathcal{T}}_K^\pi \tilde{U}_n\|_\infty^{\text{adv}} + \|\bar{\mathcal{T}}_K^\pi \tilde{U}_n - \bar{\mathcal{T}}_K^\pi V^\pi\|_\infty^{\text{adv}} \\ &\leq \bar{\varepsilon}_{\text{res}, n} + \gamma^K \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}}. \end{aligned}$$

Rearrangement proves (31). Lemma F.2, applied on the augmented closure with exact centering conditional on $\bar{s}$, gives (32). Corollary 5.3 supplies the $\bar{\tau}_\infty$ refinement. Under the exact pointwise mixture, substitution into Theorem 7.3 proves (36).

For the target-network and finite-reference statements, apply Corollary D.1 on the same augmented MDP with its target-depth, bounded-endpoint, common-boundary, frozen-policy trajectory, and exact full-state baseline hypotheses. This gives every barred substitution while keeping F_n and the deployed kernel fixed. $\square$

F.11 WEIGHTED- L_p BOUND

Proof of Proposition G.4. Let $e_m := U_m - V^\pi$. Subtracting the fixed-point identity $V^\pi = \mathcal{T}_K^\pi V^\pi$ from the definition of r_m gives

$$e_m = r_m + \beta P e_m. \quad (73)$$

Because e_m and r_m are bounded and $\beta < 1$, iteration of this identity and uniform decay of the remainder yield

$$e_m = \sum_{t=0}^{\infty} \beta^t P^t r_m, \quad |e_m| \leq \sum_{t=0}^{\infty} \beta^t P^t |r_m|. \quad (74)$$

Define the Markov kernel $Q := (1 - \beta) \sum_{t \geq 0} \beta^t P^t$. Then $\rho Q = d_\rho^{\pi, K}$ and $|e_m| \leq Q|r_m|/(1 - \beta)$. Jensen’s inequality for the Markov kernel Q gives

$$\begin{aligned} \|e_m\|_{p, \rho}^p &\leq \frac{1}{(1 - \beta)^p} \int_{\mathcal{C}} Q|r_m|^p d\rho \\ &= \frac{1}{(1 - \beta)^p} \int_{\mathcal{C}} |r_m|^p dd_\rho^{\pi, K} \\ &\leq \frac{C_{\rho, \mu}^{\pi, K}}{(1 - \beta)^p} \|r_m\|_{p, \mu}^p. \end{aligned} \quad (75)$$

Taking p th roots and applying the triangle inequality proves Eq. (87). Under Eq. (88), the same change-of-measure calculation gives $\|U_n - U_m\|_{p, \rho} \leq (C_{\rho, \mu}^{(0)})^{1/p} \|U_n - U_m\|_{p, \mu}$, which proves Eq. (89). $\square$

F.12 DEPLOYMENT PERTURBATION

Proof of Theorem E.1. Let $\mu_t^\alpha, \tilde{\mu}_t$ be the state laws and set $d_t = \text{TV}(\tilde{\mu}_t, \mu_t^\alpha)$. Their common sub-measure $\lambda_t = \tilde{\mu}_t \wedge \mu_t^\alpha$ has mass $1 - d_t$; each remaining measure has mass d_t . Appending the common reward-transition kernel K to the two action laws at a state cannot increase total variation. Their joint action-reward-successor laws therefore differ by at most δ_{dep} .

Write $J_t^\alpha, \tilde{J}_t$ for the full joint laws of state, action, reward, and successor. Integrating the statewise bound over λ_t and bounding the residual components by their mass gives

$$\text{TV}(\tilde{J}_t, J_t^\alpha) \leq d_t + (1 - d_t)\delta_{\text{dep}}.$$

Marginalization gives the same bound for d_{t+1} . Since $d_0 = 0$, induction yields $d_t \leq 1 - (1 - \delta_{\text{dep}})^t$ and

$$\text{TV}(\tilde{J}_t, J_t^\alpha) \leq 1 - (1 - \delta_{\text{dep}})^{t+1}.$$

The exponent is $t + 1$ because the reward at time t follows decision t . Both rewards lie in an interval of width Δ_r , so

$$|\mathbb{E}[\tilde{r}_t] - \mathbb{E}[r_t^\alpha]| \leq \Delta_r \text{TV}(\tilde{J}_t, J_t^\alpha) \leq \Delta_r [1 - (1 - \delta_{\text{dep}})^{t+1}].$$

Sum these reward differences for $t = 0, \dots, H - 1$ and let $H \rightarrow \infty$. Bounded rewards justify the limit, and

$$\begin{aligned} \sum_{t=0}^{\infty} \gamma^t [1 - (1 - \delta_{\text{dep}})^{t+1}] &= \frac{1}{1 - \gamma} - \frac{1 - \delta_{\text{dep}}}{1 - \gamma(1 - \delta_{\text{dep}})} \\ &= \frac{\delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma\delta_{\text{dep}})}. \end{aligned} \quad \square$$

F.13 UNIFORM-KL DEPLOYMENT BOUND

Proof of Corollary E.3. Pinsker’s inequality in either direction and $\text{TV} \leq 1$ show that the actual uniform TV discrepancy is at most $\bar{\delta}_\kappa$. Apply Theorem E.1 with $\delta_{\text{dep}} = \bar{\delta}_\kappa$; the right-hand side is increasing in this radius. If the selected KL divergence is infinite, this corollary does not apply; the TV theorem remains available when Eq. (54) is established directly. $\square$

F.14 CPI WITH DEPLOYMENT DISCREPANCY

Proof of Corollary E.4. Lower-bound $\eta(\tilde{\pi})$ by $\eta(\pi_\alpha)$ minus Eq. (55), then apply Eq. (24) with $\Xi_\alpha \leq B_\alpha$. $\square$

F.15 RESIDUAL-TO-CPI SUBSTITUTIONS

Proof of Corollary D.1. Apply Eq. (3) to $V = U_n$ to obtain Eq. (46). Lemma F.2 yields (47), and Corollary 5.3 sharpens the candidate bias to $2\gamma\tau_\infty\varepsilon_{\text{res},n}/(1 - \gamma^K)$. Substitution into Theorem 7.3 gives Eq. (48); $\tau_\infty \leq 1$ recovers the uniform fallback.

For the target-network forms, combine Proposition 6.1 with Eq. (3) to obtain Eq. (49). The triangle inequality through U_m gives Eq. (50). Corollary 5.3 applied at depth q proves Eq. (51); substitution into Theorem 7.3 gives Eq. (52). Applying the same overlap bound to the depth- n estimator and the radius in Eq. (50) gives Eq. (53). $\square$

G OPTIONAL CONTRACTION AND WEIGHTED NORMS

G.1 RECURRENT-CONTRACTION SPECIALIZATION

Assumption G.1 (Domain-relative forward-invariant contraction). Fix a declared nonabsorbing state domain $\mathcal{D}^\circ$. The latent space $\mathcal{Z}$ is a complete normed space (in our implementation, $\mathbb{R}^d$ with the Euclidean norm). For $s \in \mathcal{D}^\circ$, set

$$T_s(z) := \tilde{f}_\theta(z, y(s), x(s)).$$

There exist a closed set $\mathcal{Z}_{\text{inv}} \subseteq \mathcal{Z}$ and $L_z \in [0, 1)$ such that, uniformly for every $s \in \mathcal{D}^\circ$,

$$T_s(\mathcal{Z}_{\text{inv}}) \subseteq \mathcal{Z}_{\text{inv}}, \quad \|T_s(z) - T_s(z')\| \leq L_z \|z - z'\| \quad (z, z' \in \mathcal{Z}_{\text{inv}}).$$

Every initialization used by the result satisfies $z_{\text{init}}(x(s), y(s)) \in \mathcal{Z}_{\text{inv}}$ for every $s \in \mathcal{D}^\circ$.

For the specialization below use $\mathcal{D}^\circ = \mathcal{C}^\circ$ and finite C_z from Eq. (40). We use $L_z^0 = 1$ for every $L_z \in [0, 1)$ and $L_z^n = 0$ for $n \geq 1$ when $L_z = 0$.

Proposition G.2 (Domain-relative fixed point and finite-unrolling bias (Banach)). *Under Assumption G.1, for every $s \in \mathcal{D}^\circ$ the map T_s has a unique fixed point $z^*(s) \in \mathcal{Z}_{\text{inv}}$. For every $z^{(0)}(s) \in \mathcal{Z}_{\text{inv}}$, its iterates converge linearly:*

$$\|z^{(t)}(s) - z^*(s)\| \leq L_z^t \|z^{(0)}(s) - z^*(s)\|. \quad (76)$$

If $\mathcal{D}^\circ = \mathcal{C}^\circ$, C_z is finite as in Eq. (40), and $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$, then uniformly on $\mathcal{C}^\circ$,

$$\|z^{(n)}(s) - z^*(s)\| \leq \frac{L_z^n}{1 - L_z} C_z. \quad (77)$$

Fixed-point existence is asserted only for input pairs represented by the declared domain. Assumption G.1's metric contraction and invariance clauses alone do not imply measurability of $s \mapsto z^*(s)$.

For a sufficient measurability condition, equip the closed set $\mathcal{Z}_{\text{inv}}$ with its Borel σ -algebra and $\mathcal{D}^\circ$ with its declared σ -algebra. If

$$T : \mathcal{D}^\circ \times \mathcal{Z}_{\text{inv}} \longrightarrow \mathcal{Z}_{\text{inv}}, \quad T(s, z) = T_s(z), \quad (78)$$

is jointly measurable and $z^{(0)} : \mathcal{D}^\circ \rightarrow \mathcal{Z}_{\text{inv}}$ is measurable, then every iterate $z^{(j)}$ and the fixed-point map z^* are measurable.

For any result below that invokes L_V on a declared nonabsorbing domain $\mathcal{D}^\circ$, let $\mathcal{Z}_{\mathcal{D}}(s)$ be the set of latent arguments compared by that result at s . Assume the scalar head is uniformly Lipschitz while the instance and plan are held fixed: for every $s \in \mathcal{D}^\circ$ and all $z, z' \in \mathcal{Z}_{\mathcal{D}}(s)$,

$$|V_\psi(z, x(s), y(s)) - V_\psi(z', x(s), y(s))| \leq L_V \|z - z'\|. \quad (79)$$

For a contractive specialization one may take $\mathcal{Z}_{\mathcal{D}}(s) = \mathcal{Z}_{\text{inv}}$; a pathwise result needs the condition only at the latent pairs it displays.

Assumption G.3 (Domain-explicit bounded value heads). For an invocation on a declared nonabsorbing state domain $\mathcal{D}^\circ$, declare a latent set $\mathcal{Z}_{\text{bd}} \subseteq \mathcal{Z}_{\text{inv}}$. It is either all of $\mathcal{Z}_{\text{inv}}$ or an explicitly identified smaller set containing every latent used by that result. There exists $V_{\text{max}} < \infty$ such that, for every represented $s \in \mathcal{D}^\circ$ and every $z \in \mathcal{Z}_{\text{bd}}$,

$$|V_\psi(z, x(s), y(s))| \leq V_{\text{max}}, \quad |V_{\bar{\psi}}(z, x(s), y(s))| \leq V_{\text{max}}. \quad (80)$$

The absorber is handled by its separately declared finite scalar boundary; no absorbing latent is introduced. This assumption can supply boundedness in the optional contractive specializations, but it does not replace any explicit $B_b(\mathcal{C})$ premise in a Bellman theorem.

For the optional contractive specialization, invoke Assumption G.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the finite constant C_z above. For each represented $s \in \mathcal{C}^\circ$, write $z^*(s)$ for the unique fixed point of T_s and $U_*(s) = V_\psi(z^*(s), x(s), y(s))$ on $\mathcal{C}^\circ$. The joint measurability and measurable initialization make $s \mapsto z^*(s)$ measurable by Proposition G.2. Invoke Assumption G.3 on $\mathcal{D}^\circ = \mathcal{C}^\circ$ with $\mathcal{Z}_{\text{bd}} = \mathcal{Z}_{\text{inv}}$; together with the separate finite absorbing boundary, this makes U_* bounded on the closure. Set $U_*(s_{\text{abs}}) = -C_{\text{max}}$ and define

$$\varepsilon_{\text{res}}^* := \|U_* - \mathcal{T}_K^\pi U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (81)$$

Consecutive latent differences satisfy $\|z^{(j+1)} - z^{(j)}\| \leq L_z^j C_z$ uniformly. Hence Theorem 5.1 gives

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V C_z \frac{L_z^n (1 - L_z^{m-n})}{1 - L_z} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (82)$$

Uniform convergence $U_m \rightarrow U_*$ and Bellman contraction imply convergence of the residual norm. Taking $m \rightarrow \infty$ recovers

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \underbrace{\frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{fixed-point residual}} + \underbrace{L_V \frac{L_z^n}{1 - L_z} C_z}_{\text{geometric path bound}}. \quad (83)$$

Proofs of Proposition G.2 and Eqs. (82) and (83) appear in Appendix G.3.

G.2 A WEIGHTED- L_p FIXED-POLICY RESTATEMENT

A weighted norm can replace the population supremum when a reference measure covers the fixed policy's discounted occupancy. The following result states the corresponding change-of-measure cost. For a probability measure ν and $1 \leq p < \infty$, write $\|f\|_{p, \nu} := (\int_{\mathcal{C}} |f|^p d\nu)^{1/p}$.

Proposition G.4 (Weighted- L_p finite-reference bound). *Under the measurable fixed-policy hypotheses of Theorem 5.1, let ρ and μ be probability measures on $\mathcal{C}$, let $1 \leq p < \infty$, and take bounded measurable endpoints U_n, U_m . Set*

$$\beta := \gamma^K, \quad P := P_\pi^K, \quad r_m := U_m - \mathcal{T}_K^\pi U_m, \quad (84)$$

and define the discounted occupancy of the K -step skeleton by

$$d_\rho^{\pi,K} := (1 - \beta) \sum_{t=0}^{\infty} \beta^t \rho P^t. \quad (85)$$

This is a probability measure on $\mathcal{C}$. Suppose $d_\rho^{\pi,K} \ll \mu$ and

$$C_{\rho,\mu}^{\pi,K} := \left\| \frac{dd_\rho^{\pi,K}}{d\mu} \right\|_{\infty,\mu} < \infty, \quad (86)$$

where the norm is the μ -essential supremum. Then

$$\|U_n - V^\pi\|_{p,\rho} \leq \|U_n - U_m\|_{p,\rho} + \frac{(C_{\rho,\mu}^{\pi,K})^{1/p}}{1 - \gamma^K} \|U_m - \mathcal{T}_K^\pi U_m\|_{p,\mu}. \quad (87)$$

If also $\rho \ll \mu$ and

$$C_{\rho,\mu}^{(0)} := \left\| \frac{d\rho}{d\mu} \right\|_{\infty,\mu} < \infty, \quad (88)$$

then the endpoint discrepancy may instead be measured under μ :

$$\|U_n - V^\pi\|_{p,\rho} \leq (C_{\rho,\mu}^{(0)})^{1/p} \|U_n - U_m\|_{p,\mu} + \frac{(C_{\rho,\mu}^{\pi,K})^{1/p}}{1 - \gamma^K} \|r_m\|_{p,\mu}. \quad (89)$$

Proof in Appendix F.11. The coefficient concerns the one fixed kernel P_π^K ; the policy-sequence API/FVI comparisons are discussed in Section 2. For persistent deterministic latents, augmented occupancies can lie on graphs of zero mass under a product reference measure, so that measure need not dominate them. Choosing $\mu = d_\rho^{\pi,K}$ makes the coefficient one, but an exogenous training distribution need not have that coverage.

G.3 OPTIONAL RECURRENT CONTRACTION

Proof of Proposition G.2. For each $s \in \mathcal{D}^\circ$, Assumption G.1 makes T_s a contraction on the complete space $\mathcal{Z}_{\text{inv}}$. Banach's theorem gives its unique fixed point and Eq. (76).

Under the additional measurability premises in Eq. (78), the iterates are measurable by induction. Indeed, if $z^{(j)}$ is measurable, then $s \mapsto (s, z^{(j)}(s))$ is measurable into the product space, so $z^{(j+1)}(s) = T(s, z^{(j)}(s))$ is measurable. Banach convergence gives $z^{(j)}(s) \rightarrow z^*(s)$ pointwise. For every nonempty closed $F \subseteq \mathcal{Z}_{\text{inv}}$, continuity of distance and pointwise convergence give

$$d(z^*(s), F) = \lim_{j \rightarrow \infty} d(z^{(j)}(s), F).$$

The right side is a pointwise limit of measurable real functions, and

$$\{s : z^*(s) \in F\} = \{s : d(z^*(s), F) = 0\}.$$

The empty closed set has empty preimage. Since closed sets generate the Borel σ -algebra, z^* is measurable.

For the finite-unrolling bound, specialize to $\mathcal{D}^\circ = \mathcal{C}^\circ$ and take $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$:

$$z^*(s) - z^{(n)}(s) = \sum_{k=n}^{\infty} (z^{(k+1)}(s) - z^{(k)}(s)).$$

Contraction gives $\|z^{(k+1)}(s) - z^{(k)}(s)\| \leq L_z^k \|z^{(1)}(s) - z^{(0)}(s)\|$, so

$$\|z^*(s) - z^{(n)}(s)\| \leq \|z^{(1)}(s) - z^{(0)}(s)\| \sum_{k=n}^{\infty} L_z^k = \frac{L_z^n}{1 - L_z} \|z^{(1)}(s) - z^{(0)}(s)\|.$$

Equation (40) bounds the first increment by C_z , giving Eq. (77). $\square$

Proof of Eqs. (82) and (83). Under the $\mathcal{D}^\circ = \mathcal{C}^\circ$ invocation of Assumption G.1,

$$\|z^{(j+1)}(s) - z^{(j)}(s)\| \leq L_z^j \|z^{(1)}(s) - z^{(0)}(s)\| \leq L_z^j C_z.$$

Summing from $j = n$ through $m - 1$ yields (82). Proposition G.2 and the Lipschitz head give $\|U_m - U_*\|_\infty \rightarrow 0$. Moreover,

$$|\|U_m - \mathcal{T}_K^\pi U_m\|_\infty - \|U_* - \mathcal{T}_K^\pi U_*\|_\infty| \leq (1 + \gamma^K) \|U_m - U_*\|_\infty \rightarrow 0.$$

Taking $m \rightarrow \infty$ therefore proves (83). $\square$

H IMPLEMENTATION

H.1 COLLECTION, TARGETS, AND REPLAY

Algorithm 1 collects episodes under one frozen base policy and recurrent map. Value fitting changes the policy-independent value head; candidate fitting changes one policy head. In episodic mode the evaluator reinitializes z at each edit; persistent updates consume the stored pre-unroll latent. A target-head update uses $\bar{\psi}^+ = \tau \bar{\psi} + (1 - \tau) \psi^+$, where $\tau \in [0, 1]$ retains the old target. For a collected segment of length $\ell \leq K$, let d_{seg} indicate that its final transition terminates. The folded-reward implementation target is

$$G_{\text{fold}} = \sum_{k=0}^{\ell-1} \gamma^k r_k + \mathbf{1}\{\neg d_{\text{seg}}\} \gamma^\ell \bar{V}(\bar{s}_\ell). \quad (90)$$

Here $d_{\text{seg}} = 0$ implies $\ell = K$; early nonterminal segments are not admissible exact- K targets. On a terminal segment, the stored final reward is the formal terminal-transition reward plus γb , where $b = V^\pi(s_{\text{abs}})$. The zero padding and zero terminal bootstrap in Eq. (90) are therefore an accounting representation of the formal absorbing tail, not an additional self-loop.

Under frozen-base collection the conditional population target is $r_K^\pi + \gamma^K P_\pi^K \bar{V}$, or its full augmented-state counterpart. Proposition 6.1 accounts for propagated target lag.

Replay records the carried input and successor latents, the remaining-edit clock, the terminal cause and mask, and the behavior log probability. Terminal records carry the folded absorbing tail in the reward and are never bootstrapped. Only nonterminal transitions carry the post-unroll latent. Exact baseline evaluation uses Eq. (2) conditional on the full stored state.

Replay means, finite-population maxima, and local finite-difference estimates do not establish the population residual, target-lag, centering, or deployment bounds assumed by the theorems.

I MAPPING TO THE ORIGINAL TRM

The key differences:

1. **Training signal:** The original TRM is trained with a supervised loss; the analyzed UPI-TRM object uses checker-based rewards to construct one fixed-base candidate proposal and exact mixture.
2. **Episodic- z vs. persistent- z :** We analyze both; the original TRM is persistent- z .
3. **Halting:** The original TRM halts adaptively across its outer improvement steps; in the edit MDP that role belongs to the STOP action and the remaining budget h . The inner depth n is fixed in both settings and is treated here as an analysis parameter.